$$f_{XY}(x,y) = K(x+y)$$

$$f_Y(y) = \int_0^y K(x+y)\,dx = Ky^2/2 + Ky^2 = 3Ky^2/2$$

$$f_{XY}(x|0.5) = \frac{f_{XY}(x,y)}{f_Y(y)}\Big|_{y=0.5} = \frac{2}{3}\frac{(x+y)}{y^2} = \frac{8}{3}(x+0.5)$$

$\lambda = np$

$$P(A_k) = {}^nC_k p^k (1-p)^{n-k} = {}^nC_k \frac{\lambda^k}{n^k}\cdot\left(1-\lambda/n\right)^{n-k}$$

$$= \frac{n!}{(n-k)!\,k!}\,\frac{\lambda^k (n-\lambda)^{n-k}}{n^k}$$

$$= \lambda^k/k!\,\left(1-\lambda/n\right)^{n-k}\left(\prod_{r=1}^{k-1}(1-r/n)\right)$$

$$\downarrow$$

$$e^{(n-k)(-\lambda/n)} = e^{-\lambda} \quad \text{at } n\to\infty$$

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\,e^{-x^2/2\sigma^2} \qquad f_Y(y) = \frac{1}{\sqrt{2\pi\sigma^2}}\,e^{-y^2/2\sigma^2}$$

$$u = ax + by$$
$$v = a'x + b'y$$

$$J = \begin{vmatrix} \frac{dx}{du} & \frac{dx}{dv} \\ \frac{dy}{du} & \frac{dy}{dv} \end{vmatrix} = \begin{vmatrix} \frac{b'}{ab'-a'b} & \frac{-b}{b'a-a'b} \\ \frac{a'}{a'b-ab'} & \frac{-a}{a'b-ab'} \end{vmatrix}$$

$$\frac{a'u - av}{a'b-ab'} = y \qquad \frac{b'u-bv}{b'a-a'b} = x$$

$$= \frac{a'b-ab'}{(ab'-a'b)(a'b-ab')} = \frac{1}{ab'-a'b}$$

$$f_{u,v}(u,v) = f_{XY}(x(u,v), y(u,v))\,|J|$$

$$= \frac{1}{2\pi\sigma^2}\,e^{\left[(b'u-bv)^2+(a'u-av)^2\right]/(b'a-a'b)^2\cdot 2\sigma^2}\cdot\frac{1}{|ab'-a'b|}$$

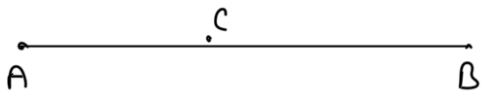
$$P(x_i = k) = p(1-p)^{k-1}$$

$$P(x_1 = k,\ x_2 = n-k)$$

$$P(X_1 \mid X_1+X_2=n) = \frac{\quad}{P(X_1+X_2=n)}$$

$$= \frac{(1-p)^{k-1} \cdot p \; (1-p)^{n-k-1} p}{\sum_{j=1}^{n-1} p^2 (1-p)^{n-2}}$$

$$= \frac{p^2 (1-p)^{n-2}}{(n-1)\, p^2 (1-p)^{n-2}} = \frac{1}{n-1}$$



$AC = x \qquad BC = 1-x \qquad y = \min(AC, BC) = \begin{cases} x & x < 0.5 \\ 1-x & x > 0.5 \end{cases}$

$$F_Y(y) = P(Y \le y) = P(\min(x, 1-x) \le y)$$

$$= P(x \le y \text{ or } x \ge 1-y)$$

$$= P(x \le y) + P(x \ge 1-y)$$

$$= y + y = 2y$$

$$\Rightarrow CDF = 2$$